



University attendance and academic performance: Encouraging student engagement

Siobhan Lucey¹ | Maria Grydaki²

¹Department of Economics, University College Cork, Cork, Ireland

²Division of Economics, University of Stirling, Stirling, UK

Correspondence

Maria Grydaki, Division of Economics, University of Stirling, FK9 4LA Stirling, UK.

Email: maria.grydaki@stir.ac.uk

Abstract

Absenteeism in higher education has motivated the investigation of its effect on academic performance. This paper examines the effect of implementing an incentive scheme on seminar *PPA* (prior preparation and attendance) and performance focusing on a cohort of international postgraduate students over two academic years in two conversion economics (quantitative and non-quantitative) modules at a Scottish business school. The results show that the scheme leads to an increase in the class *PPA* in both modules and in the probability of passing the quantitative module, however academic performance is affected at a lesser extent.

KEYWORDS

attendance policy, graduate courses, performance

JEL CLASSIFICATION

A2; A23; A29

1 | INTRODUCTION

Students' attitude of skipping classes has become a prevalent phenomenon in higher education and is a major concern for university instructors (Oldfield et al., 2018). Such non-attendance has been found to have negative consequences for both students and staff (Brauer, 1994; Landin & Perez, 2015; Sloan et al., 2020). According to Romer (1993), non-attendance in economics subjects reached high levels at three 'relatively elite' U.S. universities. He highlighted that in large classes, a greater proportion of students were skipping class than in smaller classes,

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2022 The Authors. *Scottish Journal of Political Economy* published by John Wiley & Sons Ltd on behalf of Scottish Economic Society.

fewer students were skipping classes with quantitative content than non-quantitative and more students were absent in core modules than electives.

Yet, several studies have found a positive association between students' attendance and performance (Aden et al., 2013; Dey, 2018; Marburger, 2001; Massingham & Herrington, 2006) with many of the studies specifically examining the performance of economics students (Büchle, 2021; Devadoss & Foltz, 1996; Durden & Ellis, 1995; Kirby & McElroy, 2003; Rodgers, 2001; Romer, 1993). Given the significant evidence that exists about the effect of class attendance or non-attendance on students' performance, we pursue, first, to investigate if and how an attendance policy influences class attendance, and second, to explore whether the policy affects academic performance, in terms of its impact on attendance. We contribute to the existing/U.K.-based literature as follows. First, the analysis is focused on postgraduate modules where not much research exists in this group of students and second, no study to date has analysed the relationship between an attendance policy and class attendance (and consequently academic performance) at the postgraduate level in economics modules at a business school.

There is limited previous research which focuses on postgraduate classes but at a dental school (El Tantawi, 2009). A number of studies have focused on the effect of attendance on students' performance in undergraduate modules. So, in our study, we introduce an incentive scheme to motivate higher class attendance and participation in two postgraduate conversion economics modules: one quantitative economics module and one non-quantitative economics module. These two modules are considered 'conversion' modules because students are not required to have a prior knowledge/background in either the quantitative or non-quantitative economics module. Furthermore, the cohorts of students considered in this study refer to a Scottish university and are highly international with approximately 95% of the students not coming from the United Kingdom. The latter consideration is rather important as international students make up almost a quarter of the students to Scottish institutions. Their impact is considered very positive because of the benefits to the society and additional diversity to communities they offer, broadening the learning experience at universities and supporting local businesses and jobs. Statistical data show that international students at higher education institutions (HEIs) in Scotland account for 22% of students joining HEIs compared to 19% of students studying in HEIs across the United Kingdom as a whole over the period 2016–2017 (Scottish Government, 2018).¹ Given their likely diversity in terms of (a) English language ability and (b) study methods, it is envisaged that the incentive scheme would have an additional benefit in improving the international students' academic experience (Bamford, 2008a, 2008b; Wu & Hammond, 2011). We conclude that implementing an attendance policy along the lines of the seminar incentive scheme does improve attendance but has a limited impact on academic performance. The obtained evidence applies in both quantitative and non-quantitative modules to a different extent.

Section 2 presents the related literature on the factors that affect class attendance and university performance. The introduced incentive scheme is described in Section 3. Section 4 presents the data and the econometric methodology. Section 5 reports the results. Section 6 concludes with a summary, reflections and suggestions for future research.

2 | LITERATURE REVIEW

2.1 | Attendance and academic performance

Both literature and anecdotal evidence indicate that declining student attendance is a common phenomenon at universities. Several studies have investigated university students' reasons for non-attendance. Baderin (2005) found that lack of preparation, unsuitable timing, and a preference to study on one's own as key factors. The accessibility to other sources of information (e.g. sourcing class material either online or from friends and

¹The corresponding enrollment of international students at HEIs in England, Wales and Northern Ireland is 19%, 16% and 10%, respectively.

having technology alternatives) were found to be important by Massingham and Herrington (2006) and Mearman et al. (2014). Oldfield et al. (2019) found that students' non-attendance related to their sense of belonging to the university, their views of the teaching, the fees they paid and external pressures.

Empirical studies have reported that attendance has a positive and significant impact on academic performance (Durden & Ellis, 1995; Kirby & McElroy, 2003; Romer, 1993; Stanca, 2006). Controlling for individual heterogeneity, Rodgers and Rodgers (2003) and Stanca (2006) found a positive and significant impact of class attendance on student performance in microeconomics modules at an Australian and Italian universities respectively. Cohn and Johnson (2006) obtained a similar finding using panel data for a sample of students taking principles of economics module at a U.S. university.

Fewer studies have examined the attendance–performance relationship for quantitative economic modules; notable exceptions include Rodgers (2001) who focuses the analysis on an undergraduate (UG hereafter) introductory statistics module at an Australian university; she found that attendance had a positive albeit small significant effect on grades. A similar finding is obtained by Latif and Miles (2013) for students of an introductory statistics module at a Canadian university. Moreover, Andrietti and Velasco (2015) analysed survey data collected for UG economics students enrolled in two econometrics modules offered in two consecutive terms at a Spanish university. Depending on the estimation technique used they found either a positive and significant effect of attendance on performance or an insignificant effect.

2.2 | Enhancing attendance and academic performance

Regarding the linking of attendance to module assessment, Hinnnet and Bone (2002) concluded that '[i]n students' eyes, anything that is assessed is important' (p. 54) while Baderin (2005) found that 77% of the student respondents indicated that their attitude towards tutorial attendance would change (for the better) if attendance counted towards assessment marks (p. 114). There have been various methods employed to improve attendance. Rodgers (2002) and Self (2012) discussed an incentive scheme and positive and negative attendance policies, respectively, while Dobkin et al. (2010) implemented a requirement that students who scored below the median on the midterm exam attend class. Shenker and Eyal (2013) sent a weekly email to students outlining why attendance at tutorials would be beneficial while Gbadamosi (2015) used student-led seminars to improve attendance.

Following Romer's (1993) recommendation that mandatory attendance policies should be considered by universities, several studies have attempted to measure the impact of such policies on both attendance and academic performance. Rodgers (2002) implemented an incentive scheme to encourage small-group tutorial attendance in an UG introductory statistics module at an Australian university. Under the incentive scheme, a student's grade was reduced by 1 percentage point for every tutorial missed in excess of two. While she found that the incentive scheme did increase tutorial attendance by approximately one tutorial and that there was a positive and significant relationship between tutorial attendance and academic performance for the year when the incentive scheme was in place and for the year when it was not, she concluded that the incentive scheme itself did not lead to an improvement in students' performance. Marburger (2006) explored the impact of absenteeism on performance by comparing students' exam performance in a principles of macroeconomics module in two academic years, where a university wide policy of mandatory attendance was removed in the first year and was strictly enforced in the second year. Like Rodgers (2002), his results showed that while an attendance policy did have a significant impact on reducing absenteeism, there was a weak link between absenteeism and exam performance. In contrast, when Dobkin et al. (2010) introduced a mandatory attendance policy for economic students scoring below the median on a midterm exam, they found that not only did attendance improve under the mandatory policy, but it also had a positive effect on performance, in particular for underperforming students.

In addition to looking at the impact of a mandatory policy, Self (2012) also examined whether the nature of the attendance policy was important. Using a sample of 628 students spanning eight semesters and studying



principles of macroeconomics module, she implemented three different attendance policies. One group of students had no explicit attendance policy; another group had a positive attendance policy whereby students with perfect attendance received an additional 10 credit points, those with one recorded absence received eight extra credit points and those with 2–3 recorded absences received five extra credit points. A negative attendance policy was in place for the third group whereby five credit points were deducted from the total module grade if a student missed more than 2–3 classes with deductions continually applied for additional absences. Her findings showed that (a) a mandatory attendance policy reduces absenteeism; and (b) the nature of the policy is relevant, specifically, that a negative attendance policy was more effective.

2.3 | Other factors affecting attendance and academic performance

When investigating attendance and the relationship between attendance and academic performance, several other variables are also included to account for student heterogeneity such as gender and ability. Analysis of the effects of gender on attendance find that female students miss fewer classes than male students (Arulampalam et al., 2012; Horn & Jansen, 2009; Rodgers, 2002; Ryan et al., 2010; Self, 2012; Woodfield et al., 2006). Arulampalam et al. (2012) found that females perform worse than males whereas Smith and Naylor (2001) and McNabb et al. (2002) found that women on average perform better than their male counterparts. However, Durden and Ellis (1995) and Stanca (2006) found no gender-related differences in student performance. Using attendance and performance data from an UG physiology module, Cortright et al. (2011) found a positive and significant effect between the number of classes attended for female students and their grade while there was no difference for male students. Regarding ability, various proxies have been used to capture it including grade point averages, average exam grades in previous modules and average mark on other modules in the same year/semester. Research suggests that more able students were more likely to (a) attend their modules (Andrietti, 2014; Cohn & Johnson, 2006; Rodgers, 2002) and (b) achieve higher grades in their modules (Arulampalam et al., 2012; Devadoss & Foltz, 1996).

Prior to Brexit, U.K. universities had seen increasing numbers of international students over the previous years (similar patterns are observed in other Anglophone countries such as the United States and Australia). Ryan (2011) provided an overview of the teaching and learning issues linked to the increasing number of international students choosing the United Kingdom as their study destination including the perceived problems associated with international students and the internationalisation of the curriculum. Other studies have investigated the factors affecting the academic performance of international students (Eze & Inegbedion, 2015; Lebcir et al., 2008; Rienties et al., 2012). Some have identified that many international students face language issues. Dolan and Macias (2009) noted that '[a] high proportion of international students of Economics in the U.K. report that their learning had been affected in some way due to language' [pg. 6]. Bamford (2008a, 2008b) and Lebcir et al. (2008) found that even if students met the entry requirement for language standards, they may still experience language issues due to a lack of familiarity with the technical terminology required for a specialist subject area. Meyer and Shanahan (2001) using a dataset of first-year students of Economics across two Australian universities found that students for whom English was a second language did significantly worse than those for whom English was the first language.

3 | DESCRIPTION OF THE INCENTIVE SCHEME

Previous studies suggest that students are supportive of seminar attendance linked to module assessment (Baderin, 2005; Gbadamosi, 2015; Hinnnet & Bone, 2002; Massingham & Herrington, 2006). As previously noted, various methods have been adopted to encourage attendance including the implementation of positive and

negative attendance policies (Rodgers, 2002; Self, 2012), requiring students scoring below the median on the midterm exam to attend class (Dobkin et al., 2010) and regularly contacting students by email outlining why attendance at tutorials would be beneficial (Shenker & Eyal, 2013).

For this study and for both the quantitative and non-quantitative modules, we implemented a positive seminar policy in the trial year, specifically a positive seminar prior preparation and attendance (hereafter *PPA*) policy to promote active learning. Under a positive *PPA* policy, for the quantitative module, students with perfect *PPA* in their seminars received 10% towards their final module grade, and for every seminar missed (eight seminars and eight seminar prior preparations in total) the full attendance mark of 10% was reduced by 1.25%. Similarly, for the non-quantitative module, students received 10% for full *PPA* in their seminars which were reduced by 1.67% for every seminar missed (there were six seminars and six prior preparation submissions in total).

Attendance at seminars was recorded by the tutors. Regarding the prior preparation in the trial year, seminar questions were provided to students 1 week prior to the relevant seminar and they had to submit their suggested answers in advance to the Economics administration office who maintained a record of the submissions. Given (a) the highly international cohort of students studying these modules and (b) the specialised terminology involved (particularly for the econometrics module), we envisaged that in addition to actually attending the seminars, the weekly prior preparation and submission of the seminar answers would further support students to engage with and improve both their English language skills and specialised terminology knowledge. In the control year, students were also given seminar questions in advance and strongly encouraged to do prior preparation before attending the relevant seminar (but did not count towards the final grade); however, based on tutor feedback very few students attempted the prior preparation in the control year.

The incentive scheme was explained in the module outline which was made available on the university's virtual learning environment at the beginning of the semester. It was also explained to students during the taught post-graduate (hereafter *TPG*) induction period and in the first lecture with a number of reminders also given during the semester. To evaluate the effectiveness of the incentive scheme on both attendance and performance in both modules, we compare student attendance and performance in the academic year where attendance was recorded, and the incentive scheme was in place (the Trial Year) with the previous academic year where attendance was also recorded but without an incentive scheme in place (the Control Year). With respect to the learning environment, both years were similar in terms of:

- Lecturers of the module—the same lecturers delivered the modules in both years; this controls for unobserved heterogeneity in lecturing styles.
- The content and presentation were fundamentally the same in both years.
- The method of assessment was the same in both years except for the incentive scheme. For the quantitative module, there was a mid-semester class test and an end of semester final exam, which were of the same duration and standard in both years. The non-quantitative module was assessed by two class tests during the semester, relating to the microeconomics and macroeconomics material respectively.
- Attendance was taken by the tutors at each seminar in both years. This way of recording attendance overcomes the possible attenuation bias in the estimated coefficients arising from self-reported attendance rates.

4 | DATA AND METHODOLOGY

4.1 | Data

This study takes advantage of a natural experiment in two economics modules delivered in two academic years where an explicit incentive scheme was introduced in the second year. The data used in this study were collected in two semester-one modules, an introductory econometrics module and introductory economics module

(hereafter quantitative and non-quantitative module respectively) over two consecutive academic years, 2015/16 and 2016/17. These modules were delivered to TPG students who were enrolled in a few different finance-related master's programmes at a business school in a medium-sized Scottish university; these TPG programmes can be viewed as conversion programmes and do not require a background in accounting, economics or finance but do require a first degree in any subject or an equivalent qualification. Both modules were taught over 12 weeks in the autumn semester and teaching was carried out through lectures and seminars. The whole cohort met in a lecture setting and was also split into seminar groups. Attendance was recorded by the tutors at the seminars. The datasets for both modules were constructed using the registered class lists, records of seminar submission, seminar attendance and module performance. Students who did not sit one or more elements of assessment were excluded as they did not complete the compulsory assessment. For the quantitative module, we have a total of 330 students over the 2 years, and 236 students for both years for the non-quantitative module. The students enrolled in these modules represent a very diverse group with various academic backgrounds and nationalities. The panel properties of this dataset allow us to control for unobserved heterogeneity across students that affect both attendance and performance such as ability and motivation.

In order to investigate our research questions, the following data were collected:

- (i) **Attendance (out of 100):** For the trial year, this reflects *prior preparation and attendance* (PPA) and is measured by (a) the prior submission of the seminar problem set and (b) the subsequent physical attendance at the relevant seminar session. For the control year, this is measured by physical attendance (*Attendance*) at the relevant seminar session, although the prior preparation was suggested but not rewarded as in the trial year. Each year, the tutors collected the attendance at each seminar which was then submitted to the administration office for recording. For the trial year, in addition to attendance, the submission of the solutions to the problem sets was recorded by the Economics administration office. If students missed more than 2 weeks of seminars, the administration office contacted the student to assess if there was an issue linked to the non-attendance.
- (ii) **Performance (out of 100):** academic performance is measured by the students' grade on the k^{th} assessment; each module had two pieces of assessment. More specifically, for the quantitative economics module:
 - Control Year: Mid-term exam (40% of final module grade) and end of semester exam (60% of final module grade);
 - Trial Year: Mid-term exam (35% of final module grade), end of semester exam (55% of final module grade) and PPA (10% of final module grade).

In order to avoid endogeneity concerns, average performance is constructed as 40% of midterm exam and 60% of end term exam for both control and trial years.

For non-quantitative economics module:

- Control Year: Class test 1 (50% of final module grade) and Class test 2 (50% of final module grade);
- Trial Year: Class test 1 (45% of final module grade) and Class test 2 (45% of final module grade), PPA (10% of final module grade).

In order to avoid endogeneity concerns, average performance is constructed as 50% of class test 1 and 50% of class test 2 for both control and trial years.

The class test and exams were constructed and graded by the lecturers and were based on the theory and applications discussed in the lectures and seminar. The assessment was graded anonymously, so there is no bias introduced based on high versus low attendance rates. The pass grade for both modules is 50%.

Based on previous studies, we also collected data on a number of control variables:

- (i) Incentive scheme: this is a dummy variable which takes the value of 1 for the trial year and 0 for the control year.
- (ii) Grade of other core module in the same semester: this is the students' grade (out of 100) on the other main compulsory finance-related module taken during the same semester and serves as a proxy for students' ability. We are aware that other proxies could include a students' performance score in their primary degree or their average GPA, however the authors did not get University ethical approval to use this data.
- (iii) Gender: this is a dummy variable which takes the value of 1 for a male student and zero otherwise.

4.2 | Descriptive statistics

Table A1 in Appendix A reports descriptive statistics for the main variables for both years for the quantitative and non-quantitative economics modules respectively. The sample used consists of 330 observations for the quantitative economics module and 236 for the non-quantitative economics module, with valid information on performance (*GRADE*), attendance (*ATTEND*), ability (*GRADE_CORE*), exam score for the quantitative module (*EXAM*), coursework score (*CT* for the quantitative module and *CT1*, *CT2* for the non-quantitative) and gender (*GENDER*). To obtain the difference in means between the trial and control years, separate simple regression models were estimated with each of the aforementioned variables as the outcome variables and incentive scheme as the input variable.

Table A1 in Appendix A shows that the difference in means between the trial and control years is positive and statistically significant for *ATTEND* for both modules, but negative and statistically significant for *GRADE* and the second class test, *CT2*, for the non-quantitative module only.

4.3 | Econometric specification

4.3.1 | The effect of the incentive scheme (positive attendance policy) on attendance

Class attendance (*Attend*) is considered to be a function of students' motivation, ability and gender. In order to investigate the impact of the incentive scheme and the aforementioned factors on class attendance, the following cross-sectional data structured model is estimated:

$$ATTEND_i = a_0 + a_1 SCHEME_i + a_2 GRADE_CORE_i + a_3 GENDER_i + v_i, \quad (1)$$

where the dependent variable, *ATTEND_i*, reflects the class prior preparation and attendance (*PPA*) when the incentive scheme is in place, while it represents only physical attendance when the incentive scheme is not in place, for student *i*. The independent variables denote (i) a dummy variable for the introduced incentive scheme (*SCHEME*) to motivate students to increase their attendance and participation, (ii) ability proxied by the grade achieved on another compulsory module (*GRADE_CORE*) and (iii) a dummy for gender (*GENDER*); *v_i* is the error term.

4.3.2 | The effect of the incentive scheme on performance and the probability of success

Students' successful performance is a combination of factors such as, among others, motivation, engagement with learning and teaching activities, ability and gender. In order to investigate the impact of the incentive scheme, controlling for the above factors, on overall performance, the following cross-sectional data structured model is estimated:

$$PERFORM_i = \beta_0 + \beta_1 SCHEME_i + \beta_2 GRADE_CORE_i + \beta_3 GENDER_i + v_i, \quad (2)$$

where the dependent variable, $PERFORM_i$, reflects the average module grade (measured as 40%*CT and 60%*EXAM, for the quantitative module and 50%*CT1 and 50%*CT2, for the non-quantitative) for student i . The independent variables denote (i) a dummy variable for the introduced incentive scheme ($SCHEME$), (ii) ability proxied by the grade achieved on another compulsory module ($GRADE_CORE$) and (iii) a dummy for gender ($GENDER$); v_i is the error term.

While Equation (2) captures the actual impact of the incentive scheme and the other determinants on overall performance, a probit model is estimated in order to obtain the impact of the incentive scheme and the other factors on the *probability of passing the module*. In other words, if students increase their attendance because of the implementation of the incentive scheme, what is the probability of scoring above 50%? This is based on the classification of marks across the two main categories of 'Pass' and 'Fail' following the formal classification scheme which applies at the business school (Pass: $\geq 50\%$, Fail: $< 50\%$).

Converting average performance ($PERFORM$), to a binary variable ($PERFORM_BINARY$) which takes the value of 1 if a student passes their exam and 0 if a student does not pass their exam, the estimated probit model is the following:

$$P(PERFORM_BINARY_i = 1 | SCHEME_i, GRADE_CORE_i, GENDER_i) = \Phi(\beta_0 + \beta_1 SCHEME_i + \beta_2 GRADE_CORE_i + \beta_3 GENDER_i), \quad (3)$$

where $P(PERFORM_BINARY_i)$ denotes the probability of passing the module given the independent variables as described in Equations (1) and (2), and $\Phi(\cdot)$ is the cumulative standard normal distribution function.²

4.3.3 | Further analysis of the incentive scheme on performance

To further examine whether the positive attendance policy had an impact on academic performance, we make use of the pooled structure of the data in order to get the benefit of the larger number of observations considering the grade of the two pieces of assessment per module instead of the average module grade (as in Equation (2)). The pooled model is represented as follows:

$$PERFORM_{ik} = \gamma_0 + \gamma_1 SCHEME_i + \gamma_2 GRADE_CORE_i + \gamma_3 GENDER_i + \gamma_4 MODULE_i + \varepsilon_{ik}, \quad (4)$$

where the dependent variable, $PERFORM_{ik}$, reflects the performance by student i on assessment task k ($k = 1, 2$) across both modules. The independent variables are regressors which do not vary according to the assessment task, k . Specifically, (i) $SCHEME_i$ is the dummy variable which distinguishes between students who were exposed to the incentive scheme (trial year) and those who were not (control year), (ii) $GRADE_CORE_i$ denotes a proxy for ability which captures the grade achieved on another compulsory module in the same semester, (iii) $GENDER_i$ is a dummy for

²The probit model is primarily estimated for the quantitative module because of the larger weight of the final exam (60%) towards the final module grade. The non-quantitative module is assessed by two class tests equally weighted. However, the probit model for the non-quantitative module is also estimated for consistency purposes.

gender and (iv) $MODULE_i$ is the module fixed effects and reflects a dummy variable which takes the value 1 for the quantitative module and 0 for the non-quantitative. γ_0 reflects the student specific factors which are unobservable and invariant according to the assessment task (exam or class test for the quantitative module, first or second class test for the non-quantitative) such as motivation, life satisfaction, etc., and ε_{ik} denotes the within-entity error term which varies across students and assessment task.

In order to provide information about the impact of the incentive scheme on the performance of each module specifically (intention-to-treat per module), the following interaction model is also considered:

$$PERFORM_{ik} = \gamma_0 + \gamma_1 SCHEME_i + \gamma_2 GRADE_CORE_i + \gamma_3 GENDER_i + \gamma_4 MODULE_i + \gamma_5 SCHEME_i * MODULE_i + \varepsilon_{ik}, \quad (5)$$

where all the variables are defined as in Equation (4). The interaction term $SCHEME_i * MODULE_i$ captures the relationship between the incentive scheme and the module when the scheme is in place. Although, we estimate Equations (4) and (5) using the ordinary least squares (OLS) estimation technique, a challenge we face is the endogeneity bias, which may drive our findings. A potential source of endogeneity is usually reverse causality and omitted variables due to unobserved factors not included in the model that could affect both the incentive scheme and academic performance. We use instrumental variable (IV) analysis and apply two-stage least squares (2SLS) estimation in order to get the impact of attendance on performance per se. Given that the incentive scheme directly motivates students to increase their attendance, the effect of the incentive scheme on performance comes through the increased attendance. Therefore, we replace the variable $SCHEME$ by $ATTEND$ in specifications (4) and (5) and instrument $ATTEND$ by $SCHEME$.³ $ATTEND$ reflects PPA in the trial year (prior submission of the seminar/computer lab problem set and subsequent attendance at the relevant seminar/computer lab session) and just physical attendance in the control year.

4.3.4 | Minimum detectable effect

Following the literature (Clemens & Strain, 2021; Dayioglu-Tayfur et al., 2022; Dustmann et al., 2021), to confirm the power of the above analysis, the minimum detectable effect (MDE) size for the performance outcome for the quantitative and non-quantitative modules is calculated using the formula:

$$MDE = (t_\alpha - t_\beta) * \sqrt{\frac{(\sigma_{pop})^2}{n_t} + \frac{(\sigma_{pop})^2}{n_c}} \quad (6)$$

where MDE is the minimum detectable effect which reflects the smallest possible effect that a statistical test could detect for a given Type-I error, α and Type-II error, β ; t_α , t_β denote the t-value which corresponds to the desired significance level, α and desired power of the design $(1-\beta)$; $(\sigma_{pop})^2$ shows the variance of the outcome variable; n_t , n_c refer to the sample size of the treatment and control groups respectively.⁴ The MDE, measured as the difference in outcomes between a control and an experimental group or as the proportion for the experimental group, is the minimum effect size that can be distinguished from a null effect. It is the effect size below which the researcher cannot precisely distinguish the effect from zero. The latter implies that if the level of impact of an experimental scheme is above MDE, then the experiment would be successful.

³The correlation coefficient between $ATTEND$ and $SCHEME$ is 0.123 and is highly statistically significant (p -value = 0.000) indicating that $SCHEME$ is highly correlated with $ATTEND$.

⁴We thank an anonymous reviewer for pointing out the MDE analysis.



5 | EMPIRICAL RESULTS

5.1 | The effect of the incentive scheme on attendance

Table A2 in Appendix A reports the effect of the introduced incentive scheme and the other control variables on attendance in both the quantitative and non-quantitative modules (Equation 1).

The effect of the incentive scheme is a statistically significant increase in attendance by 6% for the quantitative module and 4.8% for the non-quantitative module in the trial year. This finding clearly indicates that students were more motivated to prepare the assignments and attend the seminars in the trial year. The latter might be closely related to the reward towards the final module grade which was attached to PPA (in line with Hinnert & Bone, 2002; Baderin, 2005). Ability has a positive and significant effect on *ATTEND* in the quantitative module, suggesting that students with high marks in the other core module taken in the same semester have higher attendance. However, ability has no significant effect on attendance in the non-quantitative module. The effect of gender is negative and significant for the quantitative module suggesting that prior preparation and attendance is higher amongst female students (the effect of gender is insignificant for the non-quantitative module). The R^2 as expected for cross-sectional data is low for all models; however, the significant F statistics indicate that each model is overall statistically significant.

5.2 | The effect of the incentive scheme on performance and the probability of success

Given the finding of a positive and significant effect on attendance, Table A3 in Appendix A presents the estimated effects of the incentive scheme and the other covariates on (i) average performance (Equation 2) and (ii) the probability of scoring a pass mark in both the quantitative and non-quantitative modules (Equation 3).⁵ The further investigation of the effect of incentive scheme on academic performance considers the classification of marks across the two main categories of 'Pass' and 'Fail' following the formal classification scheme which applies at the business school (Pass: $\geq 50\%$, Fail $< 50\%$).

The results in Part A of Table A3 in Appendix A shows that the implementation of the incentive scheme led to a significant decrease in the average performance for those students who passed by 2.52 and 2.49 percentage points in the quantitative and non-quantitative module respectively. As expected, ability seems to contribute positively to the academic performance across any exam and class test mark classification in both the quantitative and non-quantitative modules respectively. As to the effect of gender, males perform poorer than females overall.

Part B of Table A3 in Appendix A shows that the implementation of the incentive scheme increases the probability of being successful within the quantitative module. More specifically, the implementation of the incentive scheme increases the probability of passing the module by 0.109 (11 percent). This finding is in line with the strand of literature which supports the positive effect of positive attendance policies on academic performance (Dobkin et al., 2010; Rodgers, 2002). In contrast, it is found that the implementation of the incentive scheme decreases the probability of passing the non-quantitative module (by 0.164). This could be related to the nature of the material (macroeconomics) examined in the second-class test (which was used to construct the binary dependent variable) and which students have found (anecdotally) to be more difficult compared to microeconomics.

⁵ For consistency purposes we estimate the probit model and the probability of passing the non-quantitative module considering the passing score of the second piece of assessment, that is the second class test. For the quantitative module the probability of passing the module is based on the passing mark of the exam, which is the second piece of assessment of the module.

It is also shown that ability affects positively the probability of passing both quantitative and non-quantitative modules by 2 and 1 percent respectively. As for gender, males seem to have lower probability of passing the quantitative module compared to female counterparts (-0.107), while gender does not have a significant effect on the probability of passing the non-quantitative module.

5.3 | Pooled OLS and IV results

Table A4 in Appendix A presents the pooled OLS and IV estimates of *SCHEME* and *ATTEND*, respectively (for Equations 4 and 5), along with other covariates by pooling the quantitative and non-quantitative modules.

The pooled OLS model (with robust standard errors) results mainly depict a negative and statistically significant relationship between *SCHEME* and performance. More specifically, when the incentive scheme is in place, students' performance is 2.4 percentage points lower, *ceteris paribus*, across both modules compared to the performance when the incentive scheme is not in place (column (1)). The same negative and statistically significant finding is obtained when the interaction term of *SCHEME*MODULE* is added in an attempt to capture the effect of the incentive scheme given a specific module (column (2)). More specifically, the implementation of the incentive scheme decreases students' performance in the quantitative module by 13.75 percentage points, while the decrease in the non-quantitative module is 4.3 percentage points, *ceteris paribus*.⁶ Therefore, the decrease in performance is greater at the quantitative module rather than the non-quantitative in the trial year. As previously mentioned, a remaining concern, however, is that our findings may be driven by endogeneity bias, thus rendering our OLS estimates biased and inconsistent. Columns (3) and (4) show the effect of attendance (*ATTEND*MODULE*), which is instrumented by the incentive scheme, on academic performance. The IV results show that attendance leads to a decrease in performance across both modules. More specifically, the decrease in academic performance is greater in the non-quantitative rather than the quantitative module by 30.14 and 4.06 percentage points respectively.⁷ Overall and in contrast to our initial hypothesis, the results show that the implementation of the incentive scheme and *ATTEND* had a negative effect on academic performance in the year when the incentive scheme was used.

These findings are in line with Rodgers (2002) who found that while the implementation of an incentive scheme did have a positive impact on attendance, it did not improve students' performance; that is increased attendance does not necessarily mean increased learning. Her finding that students performed worse at some levels when the incentive scheme was used could also be reflected in our finding that *PPA* had no significant effect on performance in the trial year. It could also be related to the nature of the attendance policy implemented, a positive attendance policy. Previous studies including Self (2012) find that negative attendance policies are more effective and that students respond more strongly to punishment than reward.

Regarding the other independent variables, the findings reveal that ability plays a positive and significant role in academic performance. The negative and significant coefficients on the *MODULE* variable reveal that students' performance in the quantitative module is lower compared to the performance in the non-quantitative module. This finding could be explained by a number of reasons: (i) late arrivals at the university due to visa issues would result in missing the introduction to statistics lectures (part of the business school's induction programme), the first module lectures and seminars/computer labs by international students; (ii) the content of the quantitative module is considered a new subject for the majority of the students and thus, it is hard for them to digest all the delivered material in the short time period leading up to the mid-term assessment; and (iii) the process of

⁷The magnitude of the effect of attendance, instrumented by the incentive scheme, on performance across students at the quantitative module is $-30.144 + 26.083 = -4.061$, while for the non-quantitative module the magnitude is -30.144 .

⁶The magnitude of the effect of the incentive scheme on performance across students at the quantitative module is $-4.252 - 12.750 + 3.252 = -13.75$.

accumulation of knowledge throughout the 12 weeks of the module delivery prepared students sufficiently for the final assessment (exam). Finally, *GENDER* is also found to have a negative and significant effect on academic performance. The latter is in line with the related literature confirming that males perform more poorly compared to female counterparts.

5.4 | Minimum detectable effect (MDE) size

In an attempt to evaluate the effect of the implementation of the incentive scheme on performance, we calculated the MDE size, considering 80% power, 5% significance level and the sample sizes for the control and trial groups of each module separately.⁸

Table A5 in Appendix A presents the MDE size for each module at 5% significance level and conditional power 80%. Any level of impact below the MDE will not be detected. Therefore, the MDE can also be considered as the level of impact below which a programme will be unsuccessful. More specifically, the difference (MDE) in performance outcomes between control and trial groups is 0.1111 and 0.1316 for the quantitative and non-quantitative modules, respectively, when we allow the proportion of difference in the trial group to be above the proportion of the control group (columns (1) and (3) for the quantitative and non-quantitative module respectively). In other words, the MDE proportion in the trial group is 0.6171 and 0.6356 compared to the proportion of the control group of 0.506 and 0.504 for the quantitative and non-quantitative modules respectively. The latter reflects that the implementation of the incentive scheme was marginally successful as the MDE proportion in the trial group was greater than the control group. This finding is in line with the probability of passing the quantitative module which increases when the incentive scheme is in place. Moreover, allowing the proportion of difference in the trial group to be lower than the proportion of the control group (columns (2) and (4) for the quantitative and non-quantitative module respectively), it is found that the difference (MDE) in performance outcomes between control and trial groups is -0.1114 and -0.1319 for the quantitative and non-quantitative modules respectively. In other words, MDE proportion in the trial group is 0.3946 and 0.3721 compared to the proportion of the control group of 0.506 and 0.504 for the quantitative and non-quantitative modules respectively. This finding shows that the incentive scheme turned out unsuccessful as the MDE proportion in the trial group was lower than the control group. The latter is in line with the negative effect of the incentive scheme on performance as detected by the OLS and IV estimation techniques.

6 | SUMMARY, DISCUSSION AND CONCLUSIONS

The increasing tendency of students to skip classes in higher education has motivated a lot of discussion and investigation towards the factors which encourage this phenomenon and measures to alleviate it. The primary objective of this paper was to examine the effectiveness of one such measure, a positive attendance policy to encourage seminar prior preparation and attendance at a typical Scottish university. Focusing on TPG students who studied two economics modules, one quantitative and one non-quantitative, we conducted the analysis for two consecutive academic years, 2015/16 (control year) and 2016/17 (trial year).

To encourage student participation with the economics material, we introduced a positive incentive scheme in the trial year and found that there was an increase in students' attendance among all students in both the quantitative and non-quantitative modules. These findings support previous research that has demonstrated a positive association between attendance policies and attendance (Marburger, 2006; Rodgers, 2002; Self, 2012; Snyder

⁸80% power means that the experiment has a chance of 80% to end up with a p -value less than 5% in a statistical test (treatment effect is statistically significant), if there is an important difference between treatments.

et al., 2014). We found in both modules that students responded to the incentive scheme and that attendance significantly increased.

A secondary objective of this paper, and related to the first, was to examine whether the attendance policy had an impact on academic performance. Similar to Marburger (2006), Dobkin et al. (2010) and Büchele (2021), we found evidence that the implementation of the incentive scheme and class attendance had a negative and significant effect on academic performance of both the quantitative and non-quantitative modules for all students when the incentive scheme was implemented. The latter is in line with Rodgers (2002), who found that the introduction of the incentive scheme and the related increase in PPA did not necessarily translate into increased performance in the trial year. However, on a positive note, our findings did indicate that students have a higher probability of passing the quantitative module when the incentive scheme was in place. Considering the international cohort of students taking such modules, the implementation of such an incentive scheme is of particular importance in supporting the learning journey of students given the potential issues/difficulties with language and new technical/econometric terminology. We also provide evidence of the positive and significant role of ability on both university attendance and academic performance in line with the related literature (Andrietti, 2014; Devadoss & Foltz, 1996; Rodgers, 2002).

While this research related to specific cohorts of students at a Scottish business school over a 2-year period, it does provide insight into potential avenues and implications for future research. The present findings show that the instructors' intention to promote active learning (through the implementation of an incentive scheme) and to motivate students to increase their university engagement and academic performance was effective to some degree but more is required. We recognise that our study is limited by lack of access to other factors, apart from gender and ability, which affect academic performance and are individual specific. There is research which relates academic performance to other factors such as life satisfaction (Astin, 1993; Rode et al., 2005). This perspective is challenging in practice for both instructors and administrators to support students with respect to life satisfaction. However, there are other elements which universities could implement in addition to attendance policies to support student attendance and performance. For example, Teixeira (2016) argues that it is insufficient for students to 'just show up' in class, they must also be actively engaged with the material. Lin (2014) suggests, in addition to standard attendance policies, daily motivational quizzes, while Chevalier et al. (2018) find that there is increased participation in online quizzes when the quizzes count towards the final grade. Love et al. (2010) also noted that the timing of feedback as well as the type of feedback (positive or negative) was found to affect student motivation for attending class and participating in extra-credit opportunities.

Finally, as previously mentioned, the two modules analysed in this paper were delivered to a cohort of students who were predominately international (non-UK). Research has shown that there are specific factors which impact on the academic performance of international students; for example English language skills (Bamford, 2008a, 2008b; Lebcir et al., 2008) and suitability of assessment methods (De Vita, 2002). Future research could account for such factors.

ACKNOWLEDGEMENTS

We thank the participants of the Chartered Association of Business Schools (CABS) Learning, Teaching and Student Experience Conference, the Irish Academy of Management (IAM) Conference and the Annual Scottish Economic Society Conference (SES), anonymous referees, as well as Dr. Dimitrios Asteriou, Prof. George Panos, Prof. Julia Darby and Prof. Claire Economidou for helpful comments. Any errors are ours.

REFERENCES

- Aden, A., Yahye, Z. & Dahir, A. (2013) The effect of Student's attendance on academic performance: a case study at Simad university Mogadishu. *Academic Research International*, 4, 409–417.
- Andrietti, V. (2014) Does lecture attendance affect academic performance? Panel data evidence from an introductory macroeconomic course. *International Review of Economics Education*, 15, 1–16.



- Andrietti, V. & Velasco, C. (2015) Lecture attendance, study time and academic performance: a panel study. *The Journal of Economic Education*, 46, 239–259.
- Arulampalam, W., Naylor, R.A. & Smith, J. (2012) Am I missing something? The effects of absence from class on student performance. *Economics of Education Review*, 31, 363–375.
- Astin, A.W. (1993) *What matters in college: four critical years revisited*. San Francisco: Jossey-Bass.
- Baderin, M.A. (2005) Towards improving Students' attendance and quality of undergraduate tutorials: a case study on law. *Teaching in Higher Education*, 10, 99–116.
- Bamford, J. (2008a) *Improving International Students' experience of studying in the UK: a case study of international student experience*. Available from: www.economicsnetwork.ac.uk/showcase/bamford_international.htm [Accessed 13th December 2018].
- Bamford, J. (2008b) Strategies for the improvement of international Students' academic and cultural experiences of studying in the UK. In: *Hospitality, leisure, sport and tourism network: enhancing series: internationalisation*. UK: Higher Education Academy, pp. 1–10.
- Brauer, J. (1994) Correspondence: should class attendance be mandatory? *Journal of Economic Perspectives*, 8, 205–215.
- Büchele, S. (2021) Evaluating the link between attendance and performance in higher education: the role of classroom engagement dimensions. *Assessment and Evaluation in Higher Education*, 46, 132–150.
- Chevalier, A., Dolton, P. & Luhrmann, M. (2018) Making it count: incentives, student effort and performance. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, 81, 323–349.
- Clemens, J. & Strain, M.R. (2021) *The heterogeneous effects of large and small minimum wage changes: evidence over the short and medium run using a pre-analysis plan*. National Bureau of economic research (NBER Working Paper No 29264).
- Cohn, E. & Johnson, E. (2006) Class attendance and performance in principles of economics. *Education Economics*, 14, 211–233.
- Cortright, R.N., Lujan, H.L., Cox, J.H. & DiCarlo, S.E. (2011) Does sex (female versus male) influence the impact of class attendance on examination performance? *Advances in Physiology Education*, 35, 416–420.
- Dayioglu-Tayfur, M., Kucukbayrak, M. & Tumen, S. (2022) The impact of age-specific minimum wages on youth employment and education: a regression discontinuity analysis. *International Journal of Manpower* (forthcoming). Available from: <https://doi.org/10.1108/IJM-02-2021-0079>
- De Vita, G. (2002) Cultural equivalence in the assessment of home and international students: a UKexploratory study. *Studies in Higher Education*, 27, 221–231.
- Devadoss, S. & Foltz, J. (1996) Evaluation of factors influencing student class attendance and performance. *American Journal of Agricultural Economics*, 78, 499–507.
- Dey, I. (2018) Class Attendance and academic performance: a subgroup analysis. *International Review of Economics Education*, 28, 29–40.
- Dobkin, C., Gil, R. & Marion, J. (2010) Skipping class in college and exam performance: evidence from a regression discontinuity classroom experiment. *Economics of Education Review*, 29, 566–575.
- Dolan, M. & Macias, I. (2009) Motivating international students: a practical guide to aspects of learning and teaching. In: *The handbook for economics lecturers, economics network*. UK: Higher Education Academy.
- Durden, G.C. & Ellis, L.V. (1995) The effects of attendance on student learning in principles of economics. *American Economic Review Papers and Proceedings*, 85, 343–346.
- Dustmann, C., Lindner, A., Schönberg, U., Umkehrer, M. & von Berge, P. (2021) Reallocation effects of the minimum wage. *The Quarterly Journal of Economics*, 137, 267–328.
- El Tantawi, M. (2009) Factors affecting postgraduate dental Students' performance in a biostatistics and research design course. *Journal of Dental Education*, 73, 614–623.
- Eze, S. & Inegbedion, H. (2015) Key factors influencing academic performance of international Students' in UKuniversities: a preliminary investigation. *British Journal of Education*, 3(5), 55–68.
- Gbadamosi, G. (2015) Should we bother improving Students' attendance at seminars. *Innovations in Education and Teaching International*, 52, 196–206.
- Hinnet, K. & Bone, A. (2002) Diversifying assessment and developing judgement in legal education. In: Burrige, R., Hinnet, K., Paliwala, A. & Varnava, T. (Eds.) *Effective learning and teaching in law*. London: Kogan Page, pp. 52–80.
- Horn, P. & Jansen, A. (2009) An investigation into the impact of tutorials on the performance of economics students. *South African Journal of Economics*, 77, 179–189.
- Kirby, A. & McElroy, B. (2003) The effect of attendance on grade for first year economics students in University College Cork. *The Economic and Social Review*, 34, 311–326.
- Landin, M. & Perez, J. (2015) Class attendance and academic achievement of pharmacy students in a European university. *Currents in Pharmacy Teaching and Learning*, 7, 78–83.

- Latif, E. & Miles, S. (2013) Class attendance and academic performance: a panel data analysis. *Economic Papers*, 32, 470–476.
- Lebcir, R.M., Wells, H. & Bond, A. (2008) Factors affecting academic performance of international students in Project Management course: a case study from a British post 92 university. *International Journal of Project Management*, 26, 268–274.
- Lin, T. (2014) Does missing classes decelerate student Progress on exam performance? Empirical evidence and policy implications. *Journal of Education for Business*, 89, 411–418.
- Love, E.G., Love, D.W. & Northcraft, G.B. (2010) Is the end in sight? Student regulation of in-class and extra-credit effort in response to performance feedback. *Academy of Management, Learning and Education*, 9, 81–97.
- Marburger, D.R. (2001) Absenteeism and undergraduate exam performance. *Journal of Economic Education*, 32, 99–109.
- Marburger, D.R. (2006) Does mandatory attendance improve student performance? *Journal of Economic Education*, 37, 148–155.
- Massingham, P. & Herrington, T. (2006) Does attendance matter? An examination of student attitudes, participation, performance and attendance. *Journal of University Teaching and Learning Practice*, 3, 82–103.
- McNabb, R., Pal, S. & Sloane, P.J. (2002) Gender differences in educational attainment: the case of university students in England and Wales. *Economica*, 69, 481–503.
- Mearman, A., Pacheco, G., Webber, D.J., Ilevs, A. & Rahman, T. (2014) Understanding student attendance in business schools: an exploratory study. *International Review of Economics Education*, 17, 120–136.
- Meyer, J.H.F. & Shanahan, M.P. (2001) A triangulated approach to the modelling of learning outcomes in first year economics. *Higher Education Research and Development*, 20, 127–145.
- Oldfield, J., Rodwell, J., Curry, L. & Marks, G. (2018) Psychological and demographic predictors of undergraduate non-attendance at university lectures and seminars. *Journal of Further and Higher Education*, 42, 509–523.
- Oldfield, J., Rodwell, J., Curry, L. & Marks, G. (2019) A face in a sea of faces: exploring university Students' Reasons for non-attendance to teaching sessions. *Journal of Further and Higher Education*, 43, 443–452.
- Rienties, B., Beausaert, S., Grohnert, T., Niemantsverdriet, S. & Kommers, P. (2012) Understanding academic performance of international students: the role of ethnicity, academic and social integration. *Higher Education*, 63, 685–700.
- Rode, J.C., Arthaud-Day, M.L., Mooney, C.H., Near, J.P., Baldwin, T.T., Bommer, W.H. et al. (2005) Life satisfaction and student performance. *Academy of Management, Learning and Education*, 4, 421–433.
- Rodgers, J.L. & Rodgers, J.R. (2003) An investigation into the academic effectiveness of class attendance in an intermediary microeconomic theory class. *Education Research and Perspectives*, 30, 27–41.
- Rodgers, J.R. (2001) A panel-data study of the effect of student attendance on university performance. *Australian Journal of Education*, 45, 284–295.
- Rodgers, J.R. (2002) Encouraging tutorial attendance at university did not improve performance. *Australian Economic Papers*, 41, 255–266.
- Romer, D. (1993) Do students go to class? Should they? *Journal of Economic Perspectives*, 7, 167–174.
- Ryan, J. (2011) Teaching and learning of international students: towards a transcultural approach. *Teachers and Teaching*, 17, 631–648.
- Ryan, M., Delaney, L. & Harmon, C.P. (2010) *Micro-Level determinants of lecture attendance and additional study-hours*. Institute for the Study of Labor (IZA Discussion paper No. 5144).
- Scottish Government. (2018) *The Impact of International Students in Scotland*. Edinburgh: Scottish government, gov.scot. Available from: <https://www.gov.scot/binaries/content/documents/govscot/publications/corporate-report/2018/03/impact-international-students-scotland-scottish-government-response-migration-advisory-committees/documents/00532586-pdf/00532586-pdf/govscot%3Adocument/00532586.pdf> [Accessed 4th June 2021].
- Self, S. (2012) Studying absenteeism in principles of macroeconomics: do attendance policies make a difference? *The Journal of Economic Education*, 43, 223–234.
- Shenker, C. & Eyal, K. (2013) *Encouraging tutorial attendance and its impact on grades: a randomised controlled trial*. Economic Research Southern Africa (Working Papers No. 332).
- Sloan, D., Manns, H., Mellor, A. & Jeffries, M. (2020) Factors influencing student non-attendance at formal teaching sessions. *Studies in Higher Education*, 45, 2203–2216.
- Smith, J. & Naylor, R.A. (2001) Determinants of degree performance in UK universities: a statistical analysis of the 1993 student cohort. *Oxford Bulletin of Economics and Statistics*, 63, 29–60.
- Snyder, J.L., Lee-Partridge, J.E., Tomasz Jarmoszko, A., Petkova, O. & D'Onofrio, M.J. (2014) What is the influence of a compulsory attendance policy on absenteeism and performance? *Journal of Education for Business*, 89, 433–440.
- Stanca, L. (2006) The effects of attendance on academic performance: panel data evidence for introductory microeconomics. *Journal of Economic Education*, 37, 251–266.
- Teixeira, A. (2016) The impact of class absenteeism on Undergraduates' academic performance: evidence from an elite economics School in Portugal. *Innovations in Education and Teaching International*, 53, 230–242.



- Woodfield, R., Jessop, D. & McMillan, L. (2006) Gender differences in undergraduate attendance rates. *Studies in Higher Education*, 31, 1–22.
- Wu, W. & Hammond, M. (2011) Challenges of university adjustment in the UK: a study of east Asian Master's degree students. *Journal of Further and Higher Education*, 35, 423–438.

How to cite this article: Lucey, S. & Grydaki, M. (2023) University attendance and academic performance: Encouraging student engagement. *Scottish Journal of Political Economy*, 70, 180–199. Available from: <https://doi.org/10.1111/sjpe.12334>

APPENDIX A

TABLE A1 Descriptive statistics

Variables	Trial year				Control year				(Trial year-control year)
	Mean	Std. Dev.	Min	Max	Mean	Std. Dev.	Min	Max	Difference in means
Part A: Quantitative economics module									
ATTEND	95.322	9.429	50.00	100.00	89.296	14.144	25.00	100.00	6.026***
GRADE	56.000	19.068	2.35	93.10	57.086	20.529	3.20	93.40	−1.086
CT	51.092	22.453	0.00	93.00	54.269	20.999	0.00	91.00	−3.177
EXAM	59.436	19.797	0.00	94.00	58.964	23.867	0.00	97.00	0.472
GRADE_CORE	59.110	15.357	19.00	94.00	59.686	14.606	20.00	93.00	−0.576
Gender	0.601	0.491	0.00	1.00	0.605	0.490	0.00	1.00	−0.004
No. of Students	163				167				
Females	65				66				
Males	98				101				
Part B: Non-quantitative economics module									
ATTEND	92.735	16.287	16.67	100.00	88.095	17.242	0.00	100.00	4.640**
GRADE	65.573	15.568	25.00	92.00	70.387	14.208	28.00	94.00	−4.814**
CT1	68.974	17.949	19.00	97.00	68.210	15.461	26.00	98.00	0.764
CT2	62.017	17.051	27.00	92.00	72.496	16.071	25.00	98.00	−10.479***
GRADE_CORE	60.308	16.051	19.00	94.00	60.935	15.556	20.00	93.00	−0.627
Gender	0.667	0.473	0.00	1.00	0.647	0.480	0.00	1.00	0.020
No. of students	117				119				
Females	39				42				
Males	78				77				

Note: Authors' calculations; ATTEND = PPA, Prior preparation and attendance, in trial year where the incentive scheme is in place, while ATTEND = Only physical attendance, in control year where the incentive scheme is not in place; GRADE = student overall performance in module; CT = class test, first piece of assessment, at the quantitative module, EXAM = final examination, second piece of assessment at the quantitative module; CT1, CT2 = first and second class tests, respectively, at the non-quantitative module; GRADE_CORE = student performance in other core module in the same semester; GENDER = 1 if the student is male and 0 if female.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

TABLE A2 Effects on attendance

ATTEND	Quantitative module	Non-quantitative module
SCHEME	6.088*** (1.305)	4.757** (2.164)
GRADE_CORE	0.122*** (0.036)	0.088 (0.058)
GENDER	-2.267* (1.290)	-3.185 (2.047)
Constant	83.369*** (2.374)	84.822*** (3.663)
Observations	330	236
F	11.76***	4.10***
R ²	0.085	0.032

Note: This table presents the results of Equation (1) for the quantitative module and non-quantitative modules. The dependent variable is Attendance (ATTEND, %) which reflects PPA in the trial year (prior submission of the seminar/computer lab problem set and subsequent attendance at the relevant seminar/computer lab session) and just physical attendance in the control year. GRADE_CORE = student performance in other core module in the same semester; SCHEME = 1 if the incentive scheme is in place and 0 otherwise. GENDER = 1 if the student is male and 0 if female. Robust standard errors in parentheses.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

TABLE A3 Effects on average performance: OLS and Probit results

	Quantitative module			Non-quantitative module		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>PERFORM</i>	All	Pass	Fail	All	Pass	Fail
PART A: OLS						
<i>SCHEME</i>	-0.552 (1.522)	-2.515** (1.172)	2.576 (2.271)	-4.345*** (1.390)	-2.490* (1.290)	-1.309 (2.101)
<i>GRADE_CORE</i>	0.958*** (0.048)	0.527*** (0.044)	0.534*** (0.096)	0.666*** (0.043)	0.490*** (0.042)	0.309*** (0.092)
<i>GENDER</i>	-4.687*** (1.519)	-2.203* (1.183)	0.188 (2.415)	-2.628* (1.454)	-1.821 (1.330)	-2.124 (2.524)
Constant	2.736 (3.472)	35.726*** (3.056)	6.685 (4.998)	31.510*** (3.178)	43.325*** (3.168)	31.289*** (4.432)
Observations	330	224	106	236	204	32
<i>R</i> ²	0.513	0.392	0.189	0.504	0.378	0.236
	Quantitative module			Non-quantitative module		
	(1)	(2)		(3)	(4)	
<i>P(PERFORM)</i>	Pass	Marginal Effects: Pass		Pass	Marginal Effects: Pass	
PART B: Probit						
<i>SCHEME</i>	0.365** (0.171)	0.109** (0.0503)		-0.887*** (0.233)	-0.164*** (0.0443)	
<i>GRADE_CORE</i>	0.066*** (0.007)	0.0198*** (0.00210)		0.048*** (0.008)	0.00861*** (0.00144)	
<i>GENDER</i>	-0.370** (0.173)	-0.107** (0.0488)		-0.131 (0.234)	-0.0230 (0.0402)	
Constant	-3.123*** (0.420)			-1.109** (0.449)		
Observations	330	330		236	236	
Pseudo <i>R</i> ²	0.265			0.247		

Note: This table presents the results of Equations (2) and (3) in Part A and B, respectively, for both the quantitative module and non-quantitative modules. In Part A, the dependent variable is the average performance and OLS results across the two main mark classification categories are reported following the formal mark classification scheme which applies at the business school; Fail<50% and Pass ≥50%. Column (1) presents the effect of incentive scheme and the other control variables on average performance without classifying the marks. Part B reflects the probability of the incentive scheme to lead to a passing score of overall performance for the quantitative module if the exam grade is above 50% (column (1)), and for the non-quantitative module if the grade of the second class test is above 50% (column (3)). Columns (2) and (4) show the associated marginal effects. *SCHEME* = 1 if the incentive scheme is in place and 0 otherwise. *GRADE_CORE* = student performance in other core module in the same semester; *GENDER* = 1 if the student is male and 0 if female. Robust standard errors in parentheses.

****p* < 0.01; ***p* < 0.05; **p* < 0.1.

TABLE A4 Effects on performance: Pooled results

	(1)	(2)	(3)	(4)
PERFORM	OLS	OLS	IV	IV
SCHEME	-2.356** (0.932)	-4.252*** (1.246)		
GRADE_CORE	0.829*** (0.029)	0.829*** (0.029)	0.874*** (0.038)	0.875*** (0.044)
GENDER	-3.911*** (0.967)	-3.891*** (0.966)	-4.995*** (1.197)	-5.286*** (1.368)
MODULE	-11.141*** (0.912)	-12.750*** (1.302)	0.313 (5.563)	-77.911* (46.777)
SCHEME*MODULE		3.252* (1.818)		
ATTEND			-11.697** (5.392)	-30.144** (15.313)
ATTEND*MODULE				26.083 (16.402)
Constant	21.316*** (2.074)	22.251*** (2.116)	49.821*** (13.829)	99.994** (40.698)
Observations	1132	1132	1132	1132
R ²	0.442	0.444	0.237	0.064
Underidentification test (<i>p</i> -value of Kleibergen-Paap rk LM statistic)			0.000	0.013
Weak identification test (Kleibergen-Paap rk Wald F statistic)			29.479	3.138
Coefficient equality test on Scheme terms (<i>p</i> -value)		0.002		
Coefficient equality test on Module terms (<i>p</i> -value)		0.000		0.215
Coefficient equality test on Attend terms (<i>p</i> -value)				0.109

Note: The results of Equations (4) and (5) are presented using pooled OLS (columns (1)–(2)) and IV (columns (3)–(4)) estimators. The dependent variable is *PERFORM* = student's grade on each piece of assessment pooling the outcomes of both modules. *SCHEME* = 1 if the incentive scheme is in place and 0 otherwise. *GRADE_CORE* = student performance in other core module in the same semester. *GENDER* = 1 if the student is male and 0 if female. *MODULE* = 1 if the module is the quantitative one and 0 if non-quantitative. The interaction term *SCHEME*MODULE* shows the relationship between incentive scheme and module. In columns (3) and (4) *SCHEME* is replaced by *ATTEND* and the former is used as an instrument for *ATTEND*. The latter refers to *PPA* in the trial year (prior submission of the seminar/computer lab problem set and subsequent attendance at the relevant seminar/computer lab session) and just physical attendance in the control year. *ATTEND*MODULE* is the interaction term when the IV approach is used. A lower *p*-value of the coefficient equality tests indicates a significant difference between the coefficients on the *SCHEME*, *MODULE* and *ATTEND* interaction terms respectively. The *Kleibergen-Paap rk Wald (F-statistic)* tests whether instruments are weak (null), with critical values varying between 5.53 and 16.38, as well as 3.63 and 7.03, in columns (3) and (4) respectively. The *Kleibergen-Paap rk LM statistic* tests whether the equation is identified, that is that the excluded instruments are 'relevant,' meaning correlated with the endogenous regressors and rejection of the null indicates that the matrix is full column rank, that is the model is identified.

Robust standard errors in parentheses, ****p* < 0.01; ***p* < 0.05; **p* < 0.1.

TABLE A5 Minimum detectable effect size

	Quantitative module		Non-quantitative module	
	(1)	(2)	(3)	(4)
	Upper	Lower	Upper	Lower
Significance level, α	0.05	0.05	0.05	0.05
Power	0.8	0.8	0.8	0.8
Total sample size, N	660	660	472	472
Size of control group, N_1	334	334	238	238
Size of trial group, N_2	326	326	234	234
MDE	0.1111	-0.1114	0.1316	-0.1319
Proportion of N_1 , P_1	0.506	0.506	0.504	0.504
Proportion of N_2 , P_2	0.6171	0.3946	0.6356	0.3721

Note: This table presents the Minimum Detectable Effect (MDE) size for the performance outcome for the quantitative and non-quantitative modules. Power shows the probability of ending up with a statistically significant treatment effect at a given significance level, α . The control group refers to the group where the incentive scheme (the treatment) was not in place, while the trial group refers to the group where the incentive scheme (the treatment) was in place. P_1 , reflects the proportion of observations of the control group with respect to the total sample size of each module. P_2 , reflects the proportion MDE in the trial group for each module. Columns (1) and (3) ((2) and (4)) show the upper (lower) direction of the effect implying that the minimum detectable proportion of the trial group would be above (lower) the proportion of the control group.